Analog Devices Inc — Aptitude Questions

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Contents

Instructions

- Each question is followed immediately by a complete, step-by-step solution.
- All intermediate arithmetic is shown; final answers are highlighted in bold.
- Units are explicit where relevant.

1. Combinatorics: 4-character password combinations

Question. For a 4-character password, calculate the total number of possible combinations for each character set below:

(a) 52 characters (upper + lower case letters)
(b) 36 characters (26 letters + 10 digits)
(c) 40 characters (custom set of 40 allowed symbols)
(d) 26 characters (lower-case letters only)

Answer.
For a password of length 4, if each position can independently be any of k characters, total combinations $= k^4$. We show numeric steps for each case.

(a) $k = 52$**:**

$$52^4 = (52^2)^2.$$

Compute stepwise:

$$52^2 = 52 \times 52 = 2704,$$

$$52^4 = 2704^2 = 2704 \times 2704 = 7{,}311{,}616.$$

Final: 7,311,616 combinations.

(b) $k = 36$**:**

$$36^4 = (36^2)^2.$$

Compute stepwise:

$$36^2 = 36 \times 36 = 1296,$$

$$36^4 = 1296^2 = 1296 \times 1296 = 1{,}679{,}616.$$

Final: 1,679,616 combinations.

(c) $k = 40$**:**

$$40^4 = (40^2)^2.$$

Compute stepwise:

$$40^2 = 40 \times 40 = 1600,$$

$$40^4 = 1600^2 = 1600 \times 1600 = 2{,}560{,}000.$$

Final: 2,560,000 combinations.

(d) $k = 26$**:**

$$26^4 = (26^2)^2.$$

Compute stepwise:

$$26^2 = 26 \times 26 = 676,$$

$$26^4 = 676^2 = 676 \times 676 = 456{,}976.$$

Final: 456,976 combinations.

2. Robots sending periodic update packets — minimum interval

Question. There are n robots. Each robot sends an update packet of size s kilobytes every T milliseconds. The available uplink bandwidth is B megabits per second. Derive the formula for the minimum allowed update interval T (in milliseconds) so the link is not overloaded. Then compute T for $n = 120$, $s = 4$ KB, $B = 12$ Mbps. (Assume 1 KB = 1024 bytes and 1 Mbps = 10^6 bits/s.)

Answer.
Derivation:
One packet size in bits:

$$\text{bits per packet} = s\ (\text{KB}) \times 1024\ \frac{\text{bytes}}{\text{KB}} \times 8\ \frac{\text{bits}}{\text{byte}}.$$

If each robot sends one packet every T seconds, packet-rate per robot is $1/T$. For n robots, total bits per second required:

$$\text{required bps} = \frac{n \cdot s \cdot 1024 \cdot 8}{T_{\text{seconds}}}.$$

To not exceed the available bandwidth B (in bits per second),

$$\frac{n \cdot s \cdot 1024 \cdot 8}{T_{\text{seconds}}} \le B.$$

Rearrange for T_{seconds}:

$$T_{\text{seconds}} \ge \frac{n \cdot s \cdot 1024 \cdot 8}{B}.$$

Convert to milliseconds: $T_{\text{ms}} = 1000 \times T_{\text{seconds}}$:

$$\boxed{T_{\text{ms}} \ge 1000 \cdot \frac{n \cdot s \cdot 1024 \cdot 8}{B}}.$$

Numeric evaluation: $n = 120,\ s = 4$ KB, $B = 12$ Mbps $= 12 \times 10^6$ bps.
Step 1: bits per packet:

$$4\ \text{KB} \times 1024 \times 8 = 4 \times 1024 \times 8 = 4 \times 8192 = 32{,}768\ \text{bits}.$$

Step 2: total bits (one packet from each robot) $= 120 \times 32{,}768 = 3{,}932{,}160$ bits.
Step 3: minimum seconds per batch:

$$T_{\text{seconds}} \ge \frac{3{,}932{,}160}{12 \times 10^6} = 0.32768\ \text{s}.$$

Step 4: convert to milliseconds:

$$T_{\text{ms}} \ge 0.32768 \times 1000 = 327.68\ \text{ms}.$$

Final: $\boxed{T_{\min} \approx 327.68\ \text{ms}}$ — updates should be at least ≈ 327.7 ms apart to avoid overloading the 12 Mbps uplink.

3. Video server uplink capacity check

Question. A video server uplink provides 50 Mbps. Each user requires a 2 Mbps video stream. If 15 users stream simultaneously, will the server uplink be sufficient? Show step-by-step calculation.

Answer.
Step 1: total required bandwidth:

$$\text{required} = 15 \times 2 \text{ Mbps} = 30 \text{ Mbps}.$$

Step 2: available uplink = 50 Mbps. Compare:

$$30 \text{ Mbps} \leq 50 \text{ Mbps}.$$

Remaining headroom:

$$50 - 30 = 20 \text{ Mbps}.$$

Final: $\boxed{\text{Yes — the uplink is sufficient. 20 Mbps spare capacity remains.}}$

4. Shopkeeper profit / number of bags

Question. A shopkeeper has a certain number of identical bags bought at 400 each. Of these bags, 40% are sold at a 20% profit, 30% are sold at a 10% loss, and the remaining 30% are sold at cost price. The shopkeeper's overall profit (net) is 480. How many bags did he have? Show stepwise calculation.

Answer.
Let total number of bags = x. Cost price per bag = 400.
Compute contribution to net profit per bag (i.e., average profit per bag):

- 40% sold at 20% profit: profit per such bag $= 0.20 \times 400 = 80$. Per-total-bag contribution: $0.40 \times 80 = 32$.
- 30% sold at 10% loss: loss per such bag $= -0.10 \times 400 = -40$. Per-total-bag contribution: $0.30 \times (-40) = -12$.
- Remaining 30% at cost price: contribution = 0.

Average (net) profit per bag:

$$\text{avg profit per bag} = 32 - 12 + 0 = 20.$$

Total profit $= x \times 20 = 480$. Solve:

$$x = \frac{480}{20} = 24.$$

Final: $\boxed{\text{24 bags.}}$

5. Log storage with compression and replication

Question. A manufacturing tool generates 450 MB of raw log data every hour. Logs are compressed to 60% of their original size (i.e., compression reduces size by 40%). For high availability, the logs are stored as 3 replicas and retained for 96 hours. What is the total storage required? Show stepwise calculation and express the final result in MB and in GB (both decimal GB where 1 GB = 1000 MB and binary GiB where 1 GiB = 1024 MB).

Answer.
Step 1: raw per hour = 450 MB.
Compressed size per hour:

$$0.60 \times 450 = 270 \text{ MB/hour.}$$

Step 2: with 3 replicas:

$$270 \times 3 = 810 \text{ MB/hour (total replicated storage).}$$

Step 3: for 96 hours:

$$810 \times 96 = 77{,}760 \text{ MB.}$$

Convert to decimal GB:

$$\frac{77{,}760 \text{ MB}}{1000} = 77.76 \text{ GB (decimal).}$$

Convert to binary GiB:

$$\frac{77{,}760 \text{ MB}}{1024} = 75.9375 \text{ GiB.}$$

Final: $\boxed{77{,}760 \text{ MB} \approx 77.76 \text{ GB (decimal)} \approx 75.94 \text{ GiB (binary).}}$

6. Round-trip average speed (harmonic mean)

Question. A train travels from A to B at 60 km/h and returns from B to A at 90 km/h. What is the average speed for the round trip? Show steps.

Answer.
For equal distances, the average speed for a round trip is the harmonic mean of the two speeds:

$$v_{\text{avg}} = \frac{2v_1v_2}{v_1 + v_2}.$$

Plugging values $v_1 = 60,\ v_2 = 90$:

$$v_{\text{avg}} = \frac{2 \times 60 \times 90}{60 + 90} = \frac{10{,}800}{150} = 72 \text{ km/h}.$$

Final: $\boxed{72 \text{ km/h.}}$

7. Sensor stream bandwidth calculation

Question. There are 8 sensors. Each sensor transmits a 512-byte packet every 50 milliseconds. What is the minimum continuous network bandwidth required (in kbps and Mbps)? Show stepwise calculation.

Answer.
Step 1: packets per sensor per second:

$$\frac{1}{0.05\text{ s}} = 20\text{ packets/s.}$$

Step 2: bytes per sensor per second:

$$512\text{ bytes} \times 20 = 10{,}240\text{ B/s.}$$

Step 3: for 8 sensors total bytes/s:

$$10{,}240 \times 8 = 81{,}920\text{ B/s.}$$

Step 4: convert to bits/s:

$$81{,}920\text{ B/s} \times 8 = 655{,}360\text{ bps.}$$

Step 5: convert to kbps and Mbps:

$$655{,}360\text{ bps} = 655.36\text{ kbps} = 0.65536\text{ Mbps.}$$

Final: $\boxed{\approx 655.36\text{ kbps } (\approx 0.65536\text{ Mbps}).}$

8. F1 telemetry — latency and capacity check

Question. In an F1 telemetry scenario each car sends a 120-byte packet every 50 ms. Two cars send simultaneously over a single-hop link of 1 Mbps. The maximum allowable network latency is 100 ms. Will the network handle both cars' telemetry without exceeding the 100 ms latency? Show stepwise calculation considering transmission time and utilization (assume negligible propagation delay and no other traffic).

Answer.
Step 1: packet size in bits:

$$120 \text{ bytes} \times 8 = 960 \text{ bits}.$$

Step 2: each car sends 1 packet per 50 ms $\Rightarrow$ 20 packets/s per car. For two cars:

$$\text{total packets/s} = 2 \times 20 = 40 \text{ packets/s}.$$

Step 3: total bits per second:

$$40 \times 960 = 38{,}400 \text{ bps} = 38.4 \text{ kbps}.$$

Step 4: link capacity = 1 Mbps = 1000 kbps. Utilization:

$$\frac{38.4}{1000} = 0.0384 = 3.84\%.$$

(very low utilization)
Step 5: transmission delay per packet:

$$\frac{960 \text{ bits}}{1{,}000{,}000 \text{ bps}} = 0.00096 \text{ s} = 0.96 \text{ ms}.$$

Step 6: worst-case queuing delay (if two packets arrive simultaneously and one must wait for the other) $\approx$ one packet transmission time ≈ 0.96 ms. So worst-case one-hop latency $\approx$ transmission + small queuing ≤ 2 ms.
Step 7: compare with allowable latency (100 ms): 2 ms $\ll$ 100 ms.

Final: Yes — the 1 Mbps link easily handles both cars; expected one-hop latency is $\ll$ 100 ms (approximately 1–2 ms in the worst case).

End of paper. — All calculations shown exactly; numbers rounded only where explicitly indicated.